

Ramakrishna Reddy Loka

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Summary

Experienced **Software Engineer** with over **6 years of expertise** in developing **secure, scalable web applications** using **C#, .NET, ReactJS, RESTful APIs, Azure, and SQL**. Skilled in **implementing robust APIs**, creating **modern, user-friendly interfaces**, and **automating cloud deployments** in enterprise environments. Proven success in **migrating legacy systems, integrating third-party services**, and **optimizing end-to-end application performance**. Committed to delivering clean, maintainable code and practicing **Test-Driven Development (TDD)** methodologies to build scalable software solutions.

Technical Skills

Languages: C#, Python, Java, TypeScript, JavaScript, SQL

Frameworks: .NET, .NET Framework, ASP.NET Core, ASP.NET MVC, Entity Framework, Node.js, ReactJS, Redux, Alteryx

Database: SQL Server, PostgreSQL, Azure Cosmos DB

Cloud & DevOps: Azure (App Services, Functions, Key Vault, Pipelines, Repos, Boards), AWS (EC2, S3, Lambda, RDS, ECS), Jenkins, Git, Ansible, CI/CD

APIs & Integration: REST APIs, Web API, Swagger/OpenAPI, Postman, Third-party APIs

Testing & Quality Assurance: NUnit, BUnit, Pytest, Jest, Veracode, Software Validation, Test Plan Development

Visualization: Power BI, Tableau, D3.js, Data Visualization Techniques

Concepts & Methodologies: Agile (SCRUM), SDLC, OOP, Data Structures, Algorithms, Version Control

Professional Experience

Graduate Research Assistant | Northern Illinois University | DeKalb, IL

Jan 2024 - May 2025

- Designed and implemented a **reactive notebook system** in Python that automatically tracks variable dependencies to deliver **reproducible and transparent computational workflows**
- Built a **scope-aware variable search engine** using **Python's AST module** to analyze code context and map variable relationships, **reducing debugging time** and improving maintainability
- Developed **interactive data visualizations** with **D3.js, TypeScript, and JavaScript**, such as dependency graphs and timeline execution traces to enhance **workflow traceability**
- Created a **new export feature** that converts reactive notebook into regular **Jupyter Notebook (.ipynb)** format, ensuring compatibility in different environments along with data science and research tools
- Integrated **Node.js services** with a **custom Python kernel** to track variable dependency computations and connect with visualization modules, **enabling real-time data synchronization** across visualizations
- Collaborated with faculty and PhD researchers to evaluate notebook performance and usability, **refining functionality through iterative feedback** from real-world research scenarios
- Implemented **CI/CD automation** and **Git-based version control**, with extensive unit testing through Pytest and Jest, ensuring high code reliability and streamlined deployment processes

Technologies: Node.js, Alteryx, TypeScript, Python, Tableau, D3.js, JavaScript, Jupyter Lab, Git, CI/CD, RESTful APIs, JSON, NumPy, Pandas, Matplotlib, Jest, Pytest, Data Visualization, HTML, CSS

Software Engineer | Experian | India

Sep 2022 – Dec 2023

- Developed and deployed **full-stack mortgage servicing applications** using **C#, .NET Core, and SQL Server**, including advanced mortgage calculators for clients Halifax, HSBC, Lloyds, Wells Fargo
- Implemented **React-based interfaces** for client-facing portals, focusing on modular design, usability, and scalability
- Led the **migration of legacy systems to .NET 6**, improving system performance by **30%** and ensuring compatibility with modern cloud infrastructure
- Designed and implemented **RESTful APIs** with **Swagger/OpenAPI**, standardizing communication between internal tools and third-party integrations to accelerate partner onboarding
- Wrote and optimized **T-SQL queries and stored procedures**, to handle high-volume mortgage data, providing business teams with instant regulatory and operational insights
- Implemented **Test-Driven Development (TDD)** using **NUnit** and automated build, testing, and deployment pipelines through **Jenkins** and **Azure DevOps**, improving release reliability and reducing production defects
- Configured **monitoring and structured logging** through **Azure Application Insights** and **Serilog**, improving performance visibility and reducing incident resolution time
- Mentored **interns and junior developers** on .NET design patterns, debugging, and best practices, improving team capability and knowledge sharing

Technologies: C#, .NET 6, ASP.NET Core, ReactJS, RESTful APIs, Swagger/OpenAPI, SQL Server, T-SQL, Azure DevOps, Jenkins, NUnit, Git, JSON, jQuery, AJAX, Postman, PowerShell, CI/CD, Azure App Services, Azure Functions, Azure Key Vault, Application Insights, Serilog, Entity Framework, Bootstrap, HTML, CSS

Software Engineer | Tata Consultancy Services (Microsoft) | India

Apr 2021 – Sep 2022

- Developed and deployed **RESTful APIs** using **ASP.NET Core (C#)** for Microsoft's internal **DevOps automation platform**, enabling real-time tracking of builds, releases, and deployment metrics
- Built responsive **ReactJS** interfaces connected to backend APIs, enabling product managers to monitor performance trends through centralized dashboards
- Designed and maintained **SQL Server reports, stored procedures, and ETL logic in Alteryx**, automating data pipelines that reduced manual reporting effort by **over 70%** and improved decision-making accuracy
- Generated and transformed **data models for Power BI dashboards**, ensuring that product managers and leadership teams had up-to-date insights into deployment performance and release quality
- Implemented **Azure Logic Apps** and automated workflows for continuous monitoring, sending real-time alerts for failed builds, latency issues, and configuration anomalies
- Configured and optimized **CI/CD pipelines in Azure DevOps**, integrating unit tests, build validation, and release approvals to enable faster and more reliable deployments
- Performed **root cause analysis** for production incidents and implemented corrective fixes, maintaining uptime and compliance with Microsoft's internal reliability standards

Technologies: C#, .NET, ASP.NET Core, ReactJS, RESTful APIs, SQL Server, Alteryx, Power BI, Azure DevOps, Azure Logic Apps, NUnit, Git, JSON, CI/CD Pipelines, Stored Procedures

Associate Software Engineer | OpenText | India

Nov 2019 – Apr 2021

- **Developed and enhanced features** in the company's **Content Management System**, focusing on **collections and process workflows** that improved content organization, user productivity, and operational efficiency
- **Managed virtualized systems** running across multiple **Windows and Linux environments**, ensuring consistent behavior, stability, and performance across versions and distributions
- Used **Ansible** and **PowerShell** to **automate environment setup, deployment, and configuration**, significantly reducing manual effort and achieving uniform builds across testing and production servers
- Optimized **SQL queries, indexing, and stored procedures**, improving database performance and reducing query response time for large content repositories
- Delivered **critical hotfixes and production issue resolutions** by debugging C# services, analyzing logs, and validating patches under tight release schedules
- Executed **unit, functionality, and integration testing** to validate system reliability and ensure high-quality feature delivery before deployment
- Partnered with **QA and DevOps teams** to verify builds, resolve release blockers, and establish rollback mechanisms for rapid recovery

Technologies: C#, ASP.NET Core, SQL Server, Linux, Ansible, PowerShell, Git, Jenkins, JSON, Entity Framework, Unit Testing, Functionality Testing, Integration Testing, Stored Procedures, Performance Optimization, Hotfix Deployment, Debugging

Engineering Intern | OpenText | India

Jun 2019 – Oct 2019

- Built **workflow automation solutions** using OpenText **AppWorks**, streamlining document routing and approval processes to reduce manual effort and turnaround time
- Debugged and resolved defects in **.NET applications**, improving reliability by analyzing logs, refining business logic, and validating fixes in test environments
- Supported the **development of backend components** for internal automation tools, contributing to improved performance and maintainability
- Performed **functionality and integration testing** to validate workflows and ensure stable releases across environments
- Created **technical documentation and configuration notes**, helping standardize deployments and enhance knowledge sharing across teams

Technologies: OpenText AppWorks, .NET MVC Framework, C#, SQL Server, Bootstrap, HTML, CSS, JavaScript, Workflow Automation, Functionality & Integration Testing, Git, Process Optimization, Documentation

Education

Northern Illinois University | Dekalb, Illinois

Jan 2024 – May 2025

Master of Science, Computer Science

Jawaharlal Nehru Technological University | India

Aug 2016 – May 2020

Bachelor of Technology, Computer Science